

Aryan Jamshidian

Personal Portfolio

Mars Habitat Robot Competition

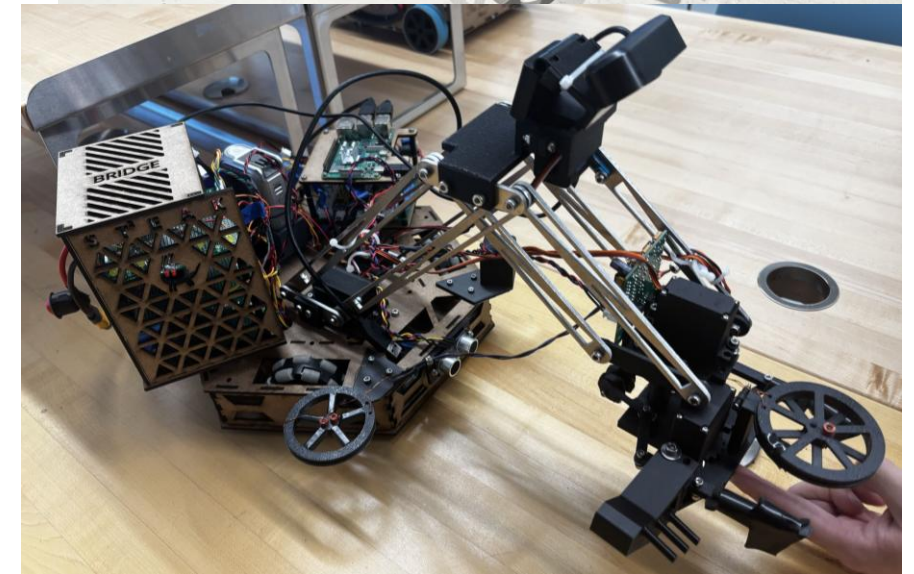
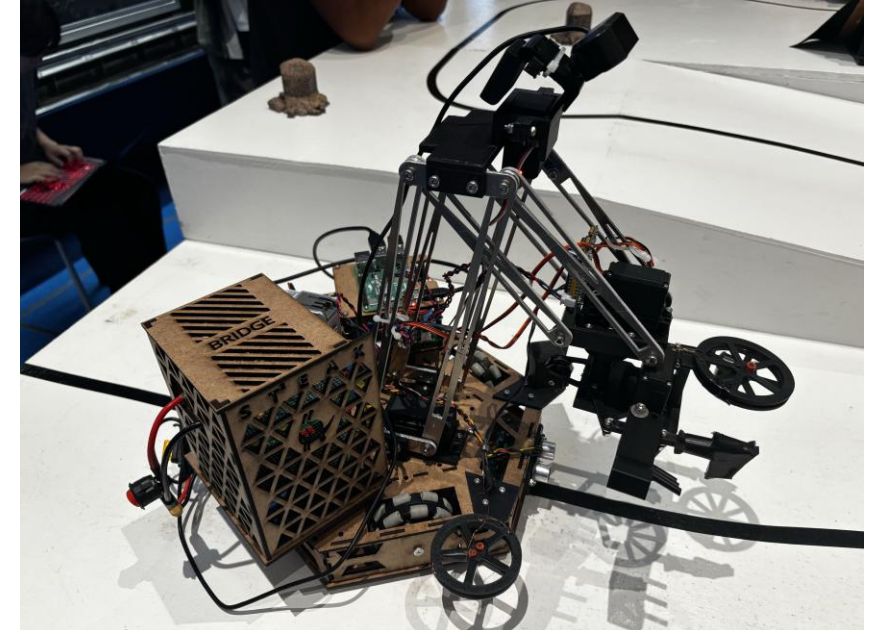
ENPH 253 – Instrument Design [Completed]

Goal:

To design a fully autonomous robot that can complete various tasks themed around Mars exploration during a timed competition.

System Overview:

- Kiwi drive (holonomic motion w/ 3 omniwheels)
- Phototransistor-based PID line following
- Sensor fusion of sonar, encoders, and cameras
- 4-DOF Arm + claw with metal detector
- ESP32 + Raspberry Pi architecture
- Computer vision using YOLO



Mars Habitat Robot Competition

ENPH 253 – Instrument Design [Completed]

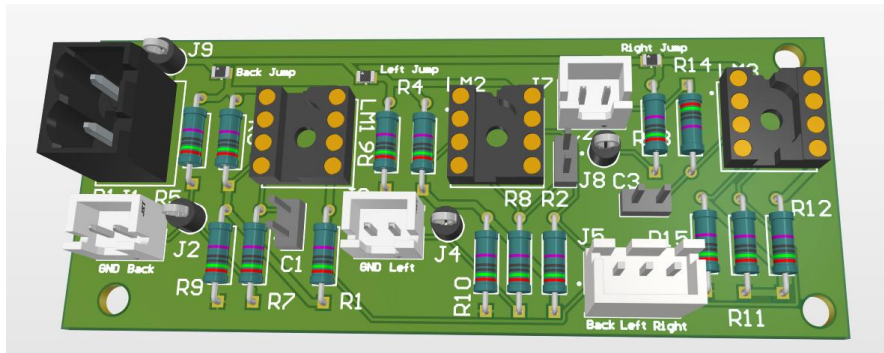
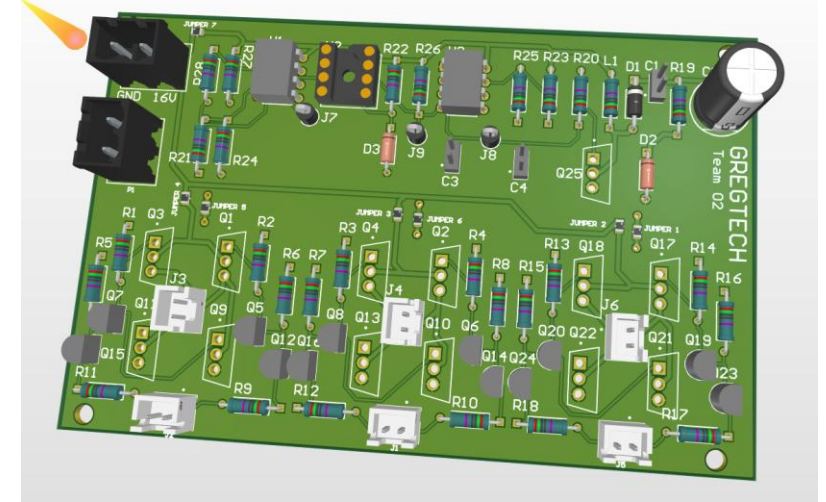
My Contributions

PCB Design:

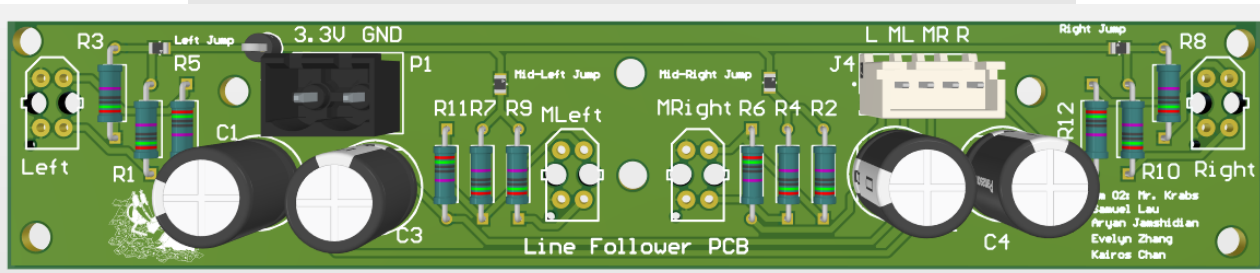
Schematics and layouts done in Altium.

All PCBs shown are custom-built for this project and soldered in-lab!

H-Bridge + Bang-Bang Controller PCB

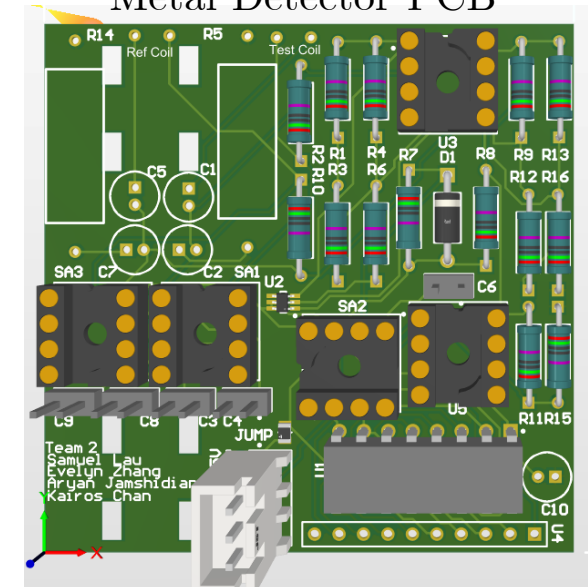


Signal Debouncer for
Rotary Encoders PCB



Phototransistor Array
Controller PCB

Metal Detector PCB



Mars Habitat Robot Competition

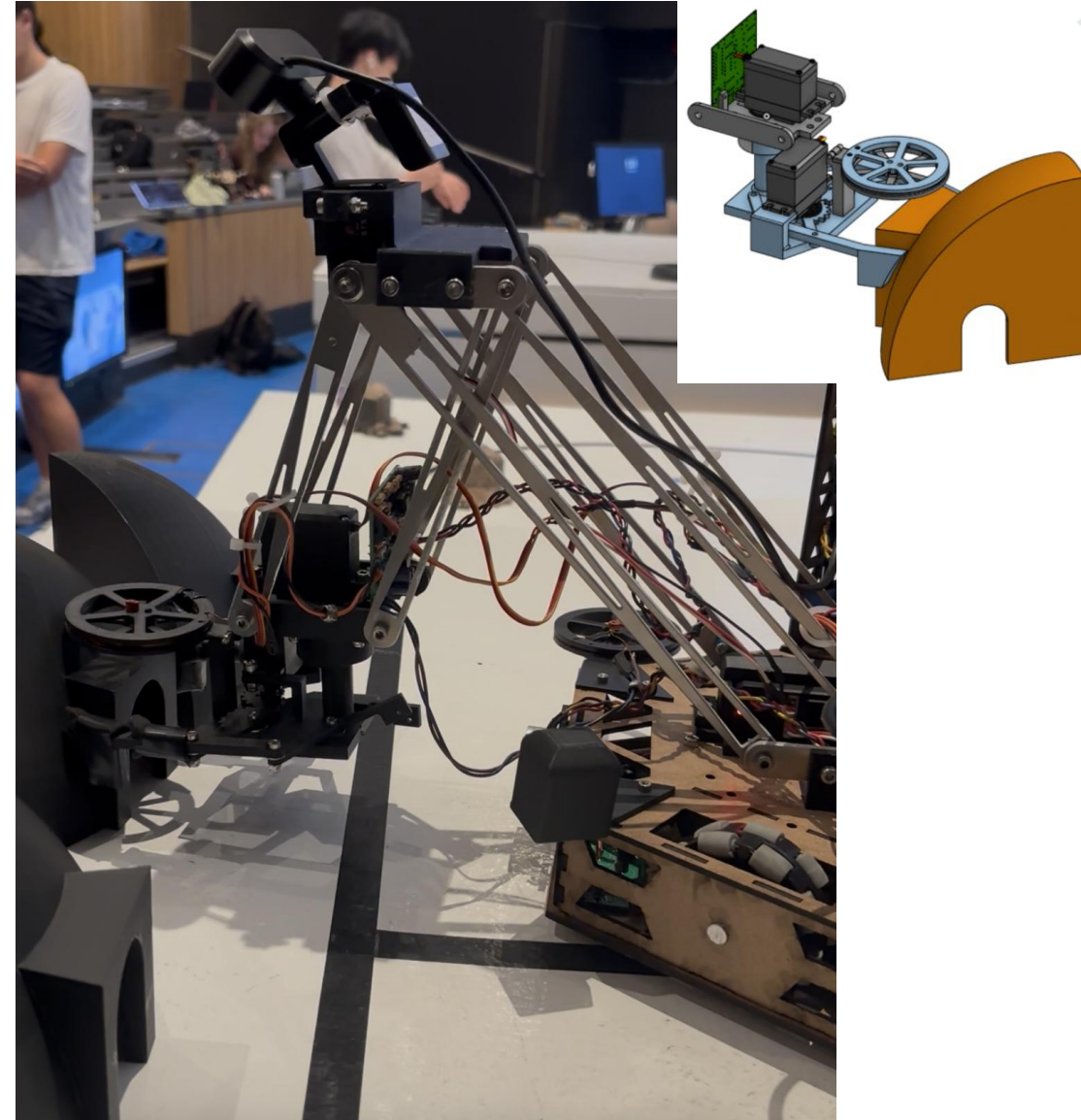
ENPH 253 – Instrument Design [Completed]

Arm:

- 4-bar linkages keep the claw parallel to the ground while maintaining mobility and reach.
- Two DOFs on the arm driven by servos, rotation of the chassis about COM acts as third DOF.

Claw:

- Has yaw + open/close DOFs, can reach around corners and deposit rock onto chassis.
- Pincer geometry is optimized for picking up rocks and aligning with the habitat door.
- Also supports the metal detector coils + PCB.

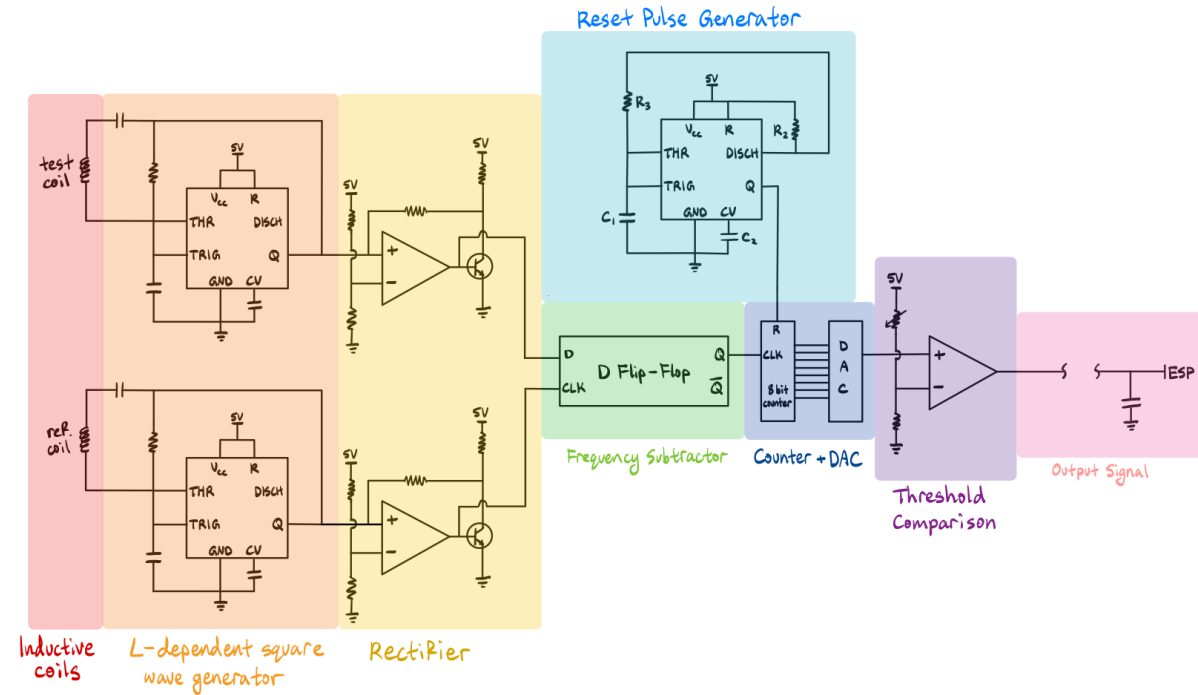


Mars Habitat Robot Competition

ENPH 253 – Instrument Design [Completed]

Metal Detector:

- I designed a custom metal detector circuit to identify aluminum foil hidden in plastic rocks.
- An inductor (made of coiled wire) will experience a decrease in inductance when exposed to ferro/paramagnetic materials due to eddy currents and polarization.
- Two matched 555 timer oscillators convert inductance into frequency. Subtracting their frequencies improves resolution, allowing for detection of paramagnets such as aluminum.
- Experimental results show that we can detect up to a 0.5% change in inductance!



All the computation and thresholding is done in hardware on the claw, insulating the system from noise as it passes back down the arm.

Electrical Architecture Redesign

UBC Rover Design Team [Completed]

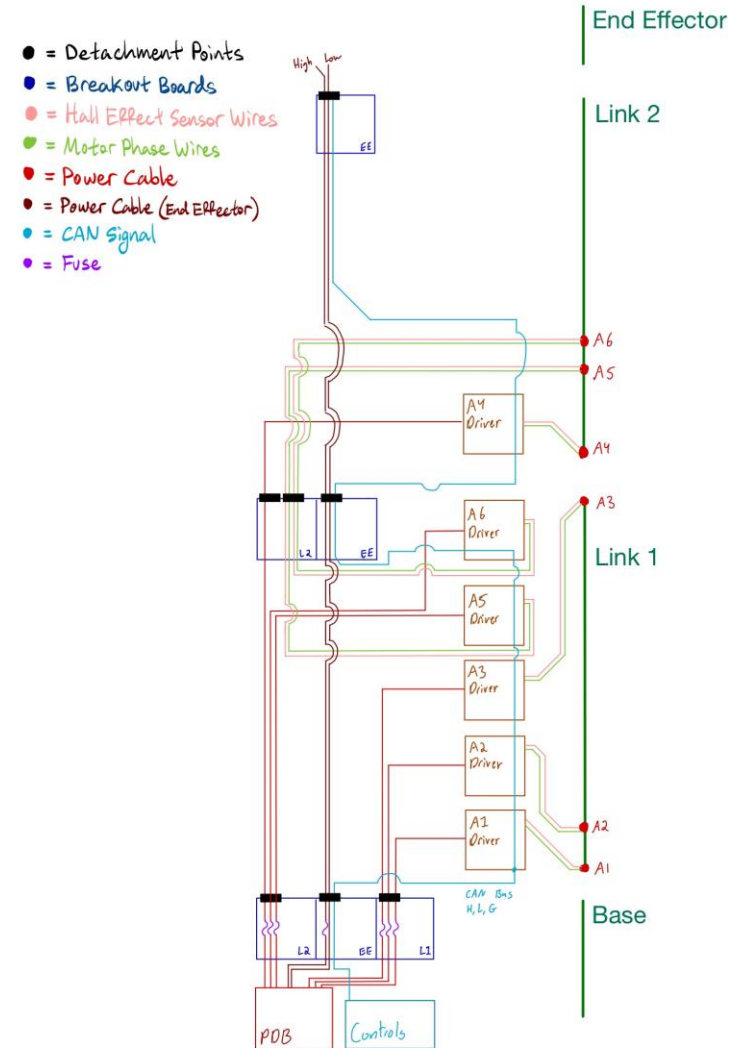
Problem:

- The rover arm has eight degrees of freedom, and extends over a metre in length.
- Running power cables directly from the chassis to every motor driver is messy, difficult to assemble, and almost impossible to test.

Solution:

- Design a new wiring architecture that creates modular disconnection points between every link, groups nearby I/O ports together, and allows for fuses, limit switches, and debugging LEDs.

Wiring Architecture Diagram

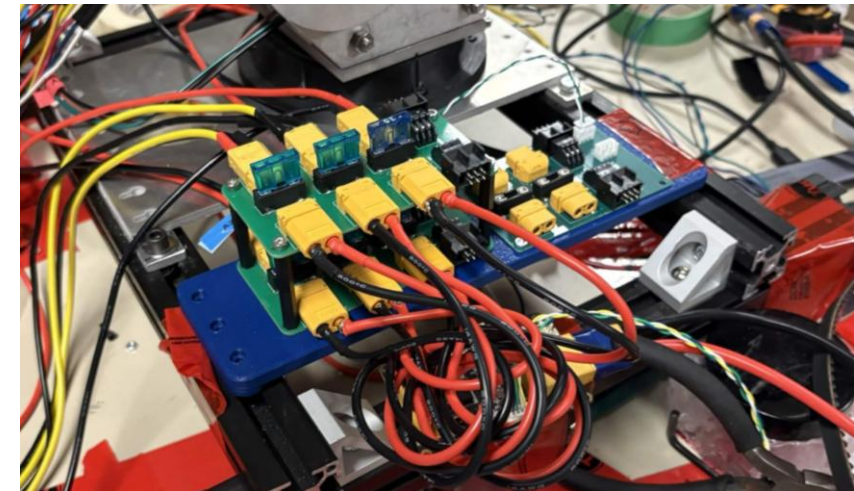
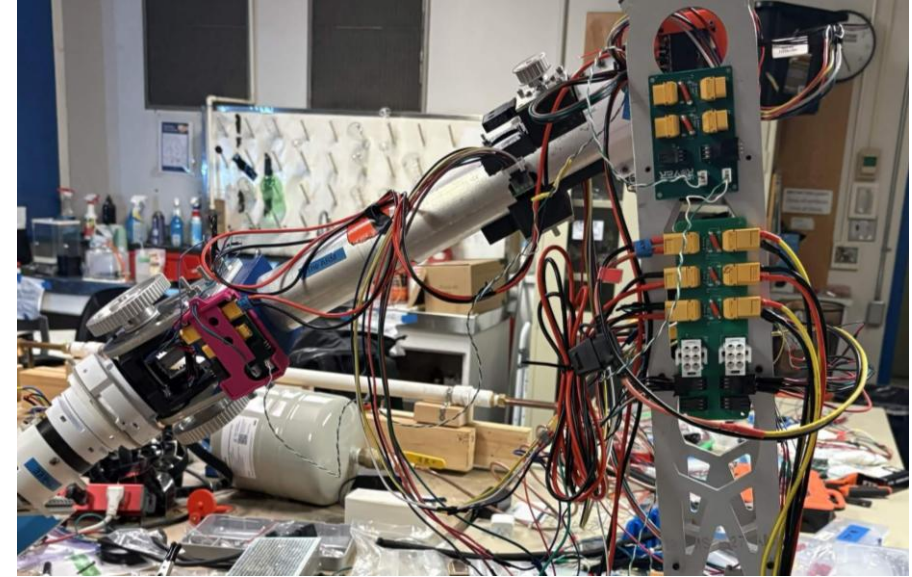


Electrical Architecture Redesign

UBC Rover Design Team [Completed]

Results:

- I designed six custom breakout PCBs in Altium, as well as PCB mounts in SolidWorks for secure and easy access.
- I used impedance matching techniques to minimize signal reflection along the CAN line.
- With the help of my team, we verified PCBs and rewired the arm.
- The project was successful, and has advanced the testing schedule by two months while improving quality of life for the arm subteam.



Silent Tokens in Transformer Models

Self-Guided Research Project [Completed]

Problem:

- The current generation of Large Language Models is very good at predicting text, but is still incapable of holding a real conversation.
- The architecture of prompt -> response -> prompt -> response places a hard restriction on the capabilities of conversational models, especially speech models.

Theory:

- LLMs process inputs in the form of tokens. Most of these tokens are words or word stubs, but some of them represent more abstract concepts such as “the start of the user prompt” or “the end of a message”.

Hypothesis:

- Transformer models could be trained to have fluid, uninterrupted conversations by adding a new token, for the concept of “silence”, into the model’s architecture.

Silent Tokens in Transformer Models

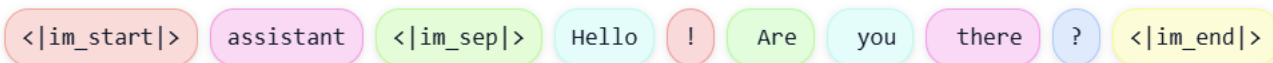
Self-Guided Research Project [Completed]

Procedure:

- I self-taught the fundamentals of machine learning and neural networks, starting from binary classifiers and progressing up to GPT-style self-attention transformers.
- I implemented and modified the GPT architecture, then retrained it on a curated dataset.
- A simplified example of my constructed training set is shown below.

Assistant: Hello! [pauses to wait for response]
Assistant: Are you there? [no reply, so speaks again]

This is a normal interaction with a person that didn't hear/understand you.

A horizontal sequence of colored rounded rectangles representing tokens. From left to right: an orange box with '<|im_start|>', a pink box with 'assistant', a light green box with '<|im_sep|>', a light blue box with 'Hello', a red box with '!', a light green box with 'Are', a light blue box with 'you', a pink box with 'there', a light blue box with '?', and a yellow box with '<|im_end|>'.

<|im_start|> assistant <|im_sep|> Hello ! Are you there ? <|im_end|>

This is what gpt-5 would be trained on.

*Since it never sees the [pause], it will learn to say
“Hello! Are you there?” without waiting for a response.

A horizontal sequence of colored rounded rectangles representing tokens. From left to right: an orange box with '<|im_start|>', a pink box with 'assistant', a light green box with '<|im_sep|>', a light blue box with 'Hello', a red box with '!', a grey box with '<|silence|>', a light green box with 'Are', a light blue box with 'you', a pink box with 'there', a light blue box with '?', and a grey box with '<|silence|>'.

<|im_start|> assistant <|im_sep|> Hello ! <|silence|> Are you there ? <|silence|>

This is what my model would be trained on.

Silent Tokens in Transformer Models

Self-Guided Research Project [Completed]

- In theory, the model would learn that after completing a thought, it should output silence tokens (which are relayed to the user as pauses).
- If the user responds, it continues as normal. But if the user doesn't respond, then after a few silence tokens have built up it will learn to start messaging again.
- The number of silence tokens allow it to differentiate between a user responding quickly or slowly.

Results:

- Unfortunately, this experiment did not yield significant results.
- Although the model did sometimes output silence tokens, they did not match the patterns from the training data and mapped to pseudo-random pauses in speech.
- I believe that this idea could be promising with a more suitable model architecture and better training set. I intend to revisit this project in the future.

Contact

Thank you for your interest!

If my experience aligns with your company goals, feel free to contact me with:

Email: ajamshidian22@gmail.com

LinkedIn: <https://www.linkedin.com/in/aryan-jamshidian-444007337/>

Phone: 604-716-4807

Or learn more about the Engineering Physics program at <https://www.engphys.ubc.ca/>